

STAT8020 Group Project

Contrarian Strategy to Beat Buy-and-Hold



Team Member:

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I. Introduction

The semi-strong-form Efficient Market Hypothesis states that stock prices reflect all publicly available information, and it is not possible to earn excess returns based on fundamental and technical analysis. However, this may not always hold in reality as investors may not always make rational investment decisions. Represented by Baron Rothschild and Warren Buffett, contrarian investors usually make good investments at cheap price when most market participants are selling off and demonstrating crowded behavior. In the US, from 1945 to 2016, there were 21 correction periods (defined as a period during which the stock market drops by 10% – 19.9%) in total, and on average, they only lasted 5 months and the market only took 4 months to recover from the bottoms. Market selloff is usually associated with increased volatility. We would like to examine a trading algorithm that uses volatility to gauge the timing of entering into and exiting from a long position in the stock market.

The Chicago Board Options Exchange publishes the CBOE Volatility Index, or VIX, which measures the implied volatility from S&P 500 index options. We have gathered a dataset from Bloomberg which contains the daily open and closing price of VIX and S&P 500 index as well as their spot month futures contracts. As shown in the following graph, VIX index has a negative correlation with S&P. VIX index tends to increase significantly during market selloff, and decrease when S&P 500 index continues to recover. Therefore, it may be profitable to long S&P 500 index futures when VIX index increases above a threshold, and close the position when VIX index remain below the threshold for a certain number of days, which is the basic idea behind our algorithm.



Plot1: Plot of VIX future and SPX future from 2008 Jan to 2018 March

II. Proposed strategy

2.1 Based on the negative correlation between VIX and the market (S&P), we propose the following strategy.

Let: VIX at open is VOpen and SPX futures at open is SOpen.

We define x as the threshold of VIX index to trigger buy and sell signal. The range of x is from 0 to 100, as we observed from the Plot 1.

Buy signal: if $V_{Open} > x$, buy SPX futures at S_{open} .

Let y is the number of consecutive days which $V_{Open} < x$

Hold: Given buy signal is triggered, $V_{Open} > x$ or $V_{Open} < x$ for less than y days consecutively

Sell signal: $V_{Open} < x$ and the stocks have been held more than y days.

The R codes for our algorithm is attached at the end.

2.2 Rationale

As mentioned in the background, the buy signal that we use is when market starts to panic because we believe that this offers attractive valuation of risk asset compared to the days when the market is less volatile. The strategy then holds the risk asset during the period of volatile time but does not immediately sell when the volatility disappears. Rather, the strategy waits for a pre-defined number of quiet days before it triggers the sell signal. This is essentially to wait for the market mood to swing to the risk-taking mode.

III. Backtesting

We backtest using VIX at open and SPX futures at open from Jan 2, 2008 to Mar 29, 2018. The backtesting methodology is as followed:

2.3.1 Given predefined holding period y , $y=10$ days until 300 days, we pick the high enough return that beats BAH for three time horizon categories: short term, medium term, and long term.

We define less than 100 trading days as short term, 100-200 trading days as medium term, 200-300 trading days as long term

2.3.2 Based on these holding period selection, we find the threshold of VIX that yields high enough return that beats BAH.

2.3.3 Note that we may select more than one threshold to be tested in out-sample testing to see its performance in out-sample period

2.3.4 Parameter selection

Based on the data during backtesting period, we observe the following result.

Number of "quiet" days before selling	Maximum profit	Number of "quiet" days before selling	Maximum profit
10	1196.96	160	1032.92
20	1260.02	170	1033.58
30	1226.44	180	1058.88
40	1330.78	190	1051.2

50	1215.4	200	1051.98
60	1132.34	210	1051.75
70	1240.66	220	1119.16
80	1235.73	230	1177.1
90	1274.7	240	1187.82
100	1367.08	250	1180.85
110	1399.26	260	1121.2
120	1340.95	270	1158.27
130	945.76	280	1220.21
140	1038.22	290	1263.36
150	1030.49	300	1341.19
Buy and Hold	1146.44		

Table 1: Backtesting Result for the Selection of Holding Period y

Note that using backtesting period data, we can see that holding period of 40 days, 110 days, and 300 days maximizes the profit and all of them beats buy and hold strategy.

Based on this observation, we then hold the holding period as those numbers and select the threshold. Note that we do not always have to pick the one that yields the maximum return as long as the return is sufficiently better than the buy-and-hold and makes sense. For example, trading only when VIX is 30 is by domain knowledge not a good idea given it is very rarely that VIX hits that number, so it might yield the maximized profit only with the given training data set.

Simulated profits for 40 “quiet” days holding period (index represents threshold)

[1] 0.00 0.00 0.00 0.00 0.00 0.00
[7] 0.00 0.00 0.00 0.00 0.00 **1330.78**
[13] 1179.03 1164.43 1068.40 1116.61 769.56 650.97
[19] 843.28 773.98 694.43 641.95 659.63 449.51
[25] 572.76 932.57 653.85 619.54 625.78 152.36
[31] 115.31 -72.82 -66.60 -66.60 -73.48 -172.23
[37] -123.73 -126.87 -125.99 -110.41 -140.00 -89.54
[43] -145.63 -154.62 -154.62 -57.08 -156.38 -153.65
[49] -153.65 -178.39 -200.75 -111.49 -154.96 -154.47
[55] -154.47 -154.47 -120.18 -34.07 -56.22 -64.54
[61] -64.54 -64.54 -64.54 -56.62 -56.62 -67.06
[67] -90.59 -90.59 -90.59 -121.35 -52.12 -34.54

[73]	-34.54	-34.54	-34.54	-34.54	-34.54	-34.54
[79]	-34.54	83.90	0.00	0.00	0.00	0.00
[85]	0.00	0.00	0.00	0.00	0.00	0.00
[91]	0.00	0.00	0.00	0.00	0.00	0.00
[97]	0.00	0.00	0.00	0.00		

Simulated profits for 110 “quiet” days holding period (index represents threshold)

[1]	0.00	0.00	0.00	0.00	0.00	0.00
[7]	0.00	0.00	0.00	0.00	0.00	0.00
[13]	0.00	1399.26	1379.51	1334.63	933.78	957.96
[19]	1074.06	915.88	686.27	564.24	420.42	428.52
[25]	454.52	557.20	494.47	557.56	597.35	131.46
[31]	100.85	144.50	145.45	145.45	127.28	162.77
[37]	263.56	266.40	268.18	287.95	254.59	287.51
[43]	252.95	251.54	251.54	345.71	71.21	-45.37
[49]	-45.37	-63.60	-57.77	-57.77	-78.96	-78.47
[55]	-78.47	-78.47	-79.80	6.31	7.36	-9.55
[61]	-9.55	-9.55	-9.55	8.21	8.21	-2.23
[67]	-6.65	-6.65	-6.65	-37.41	31.82	4.93
[73]	4.93	4.93	4.93	4.93	4.93	4.93
[79]	4.93	123.37	0.00	0.00	0.00	0.00
[85]	0.00	0.00	0.00	0.00	0.00	0.00
[91]	0.00	0.00	0.00	0.00	0.00	0.00
[97]	0.00	0.00	0.00	0.00		

Simulated profits for 300 “quiet” days holding period (index represents threshold)

[1]	0.00	0.00	0.00	0.00	0.00	0.00
[7]	0.00	0.00	0.00	0.00	0.00	0.00
[13]	0.00	0.00	0.00	0.00	1341.19	
[19]	1341.19	1341.19	1277.52	991.42	809.98	846.39
[25]	889.60	960.22	682.90	688.27	688.27	436.83
[31]	436.83	199.18	199.18	199.18	159.11	314.97

[37] 466.09 466.09 472.00 472.00 481.03 481.03
 [43] 460.79 472.97 472.97 645.10 314.04 81.18
 [49] 81.18 99.59 98.79 98.79 114.19 114.68
 [55] 114.68 114.68 121.58 207.69 203.18 193.83
 [61] 193.83 193.83 193.83 164.79 164.79 154.35
 [67] 184.50 184.50 184.50 153.74 222.97 226.39
 [73] 226.39 226.39 226.39 226.39 226.39 226.39
 [79] 226.39 344.83 0.00 0.00 0.00 0.00
 [85] 0.00 0.00 0.00 0.00 0.00 0.00
 [91] 0.00 0.00 0.00 0.00 0.00 0.00
 [97] 0.00 0.00 0.00 0.00

IV. Out-Sample Testing

Based on the back testing results, we select three sets of parameters for the out-sample testing.

- (1) Threshold VIX= 12pt , Holding periods = 40 trading days
- (2) Threshold VIX = 14pt, Holding periods = 110 trading days
- (3) Threshold VIX = 17pt, Holding periods = 300 trading days

As our holding periods is relatively long, we will select the last 250 trading days from 2017-Apr-03 to 2018-Mar-29 as our out-sample period. We will compare our cumulative profits during this period with Buy-and-Hold strategy.

We summarized our cumulative profits in out-sample period as below.

Parameter set	Cumulative profits
1. 12pt,40d	0
1a. 13pt, 40d	434.7 (+72.8%)
2. 14pt,110d	501.3 (+99.2%)
3. 17pt,300d	0
BAH	251.6

Table 1 Comparison of Trading strategies and Buy-and-Hold

It is noted that for the parameter set of (12pt, 40d) and (17pt,300d), no sell signal exists during the out-sample period. So we slightly change the threshold of 1st set parameter from 12pt to 13pt for comparison.

- (1) Threshold VIX= 13pt , Holding periods = 40 trading days

We start our long position in the first trading day (i.e. 2017-Apr-03) as VIX future opens at 13.3pt which is higher than the threshold. We establish our long position at 2358.8 pt. During the testing

period, there are two times that VIX is lower than 13pt for a consecutive 40 trading days as shown below. So we sold on 2017-Nov-02 for 2571.9pt and 2018-Jan-17 for 2785.9pt, respectively. While we generate 213.1pt and 221.6pt for each transaction.

Table 2 Count of Holding Periods to trigger sell signal

(2) Threshold VIX = 14pt, Holding periods = 110 trading days

We start to hold our long position from 2017-Apr-10 to 2018-Jan-30. We buy at 2352.9pt and sell for 2854.2pt, making a profit of 501.3pt. Later on 2018-Feb-05, another buy signal is triggered while the sell signal is not present till the end of the period.

[illegible]

Table 3 Profits table for (14pt, 110days)

(3) Threshold VIX = 17pt, Holding periods = 300 trading days

The long position is established on 2018-Feb-06 as VIX future is first-time passed over the threshold of 17pt. The opening of SPX future is 2626.6pt on that day. The VIX is continuously higher than the threshold afterwards for a period of 30 days. We can't see the performance of this strategy as the selling signal is not generated during the out-sample period. But this is a very long-term strategy which requires the VIX lower than 17pt for continuously 300 trading days.

V. Observations and Discussions

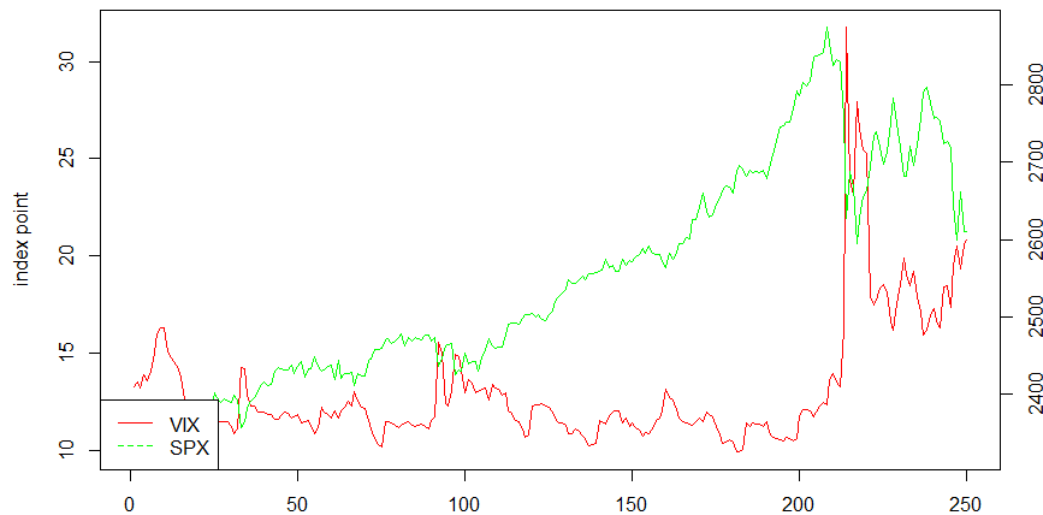
We observe that the optimal parameters obtained from the back-testing have various performance in the out-sample testing. The medium-term strategy can still render higher profits than BAH while the short-term and long-term strategy cannot be assessed due to the short time duration of out-sample period. So we try to explore the reasons behind.

(1) Changing market volatility

From the Plot 1, we can clearly see that the market is highly volatile during 2008-2011. Afterwards, it is rising stably and VIX stays low. One pre-requisite for the success application of our contrarian strategy is the high volatility and low volatility appear alternately, so that buy signal and sell signal could be generated. This is the case in the back-testing period from 2008 – 2018, while in the out-

sample period, as shown in Plot 2, the VIX starts to be low and stable, but spike up at the end of the period.

In addition, the parameters (e.g. threshold, holding period) are highly dependent on the volatility structure. So when the volatility structure changes dramatically, the parameters should be adjusted accordingly.



Plot 2 Plot of VIX future and SPX future in the Out-sample period 2018

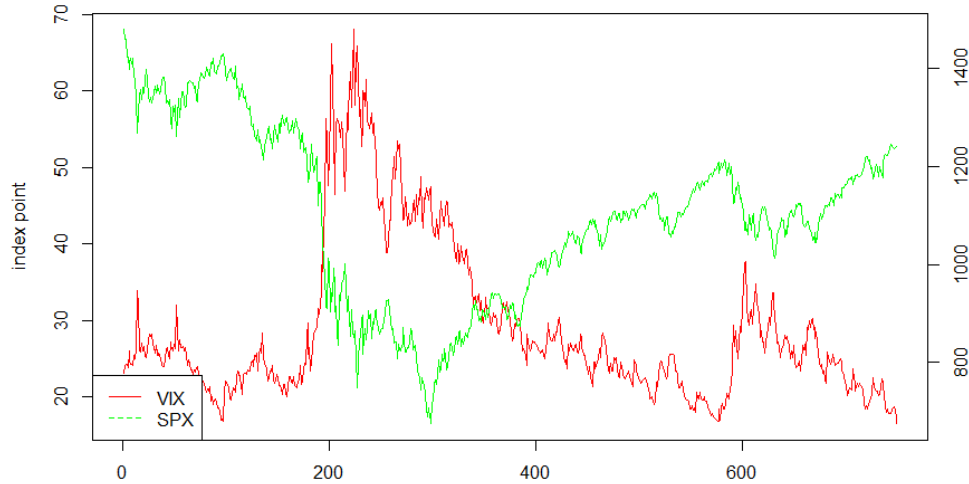
(2) Trading Frequency and Holding Period

As mentioned in the Part 3 - Description, this is a conservative contrarian strategy to buy low and sell high. As we assume the investor is so conservative that only if the low market volatility last for a consecutive period, a sell signal would be generated. If the high and low volatility cluster oscillate very frequently, a sell signal will not be generated and we will hold the SPX future for a significant long time. As a result, this strategy is more suitable to investors who prefer the long-term holding than the short-term trading.

(3) Further Improvements

1. Separate buy and sell threshold

We may consider to set different VIX thresholds for buy and sell signal to enhance the robustness of the model in fitting different market conditions. For example, when the market is volatile, we may want to enter the market when the SPX is low. If we set our buy threshold the same with the sell threshold, say 20, we will buy SPX future around 60. However, if we set the buy threshold at 60, we can buy SPX future at 30, which is significantly lower than the previous purchase price.



Plot 3 Plot of VIX future and SPX future of 2008-2010

2. Selections of out-sample period

We present the performance of our strategy in different out-sample periods lasting for 1 year, 2 years and 5 years as below.

Parameter set	2017-Apr-03 to 2018-Mar-29	2016-Apr-06 to 2018-Mar-29	2013-Jan-03 to 2018-Mar-29
1. 13pt,40d	434.7 (+183.1)	751.8 (+183.1)	1352.4 (+183.1)
2. 14pt,110d	501.3 (+249.7)	812.5 (+243.8)	1413.1 (+243.8)
3. 17pt,300d	0	771.7 (+203)	1192.5 (+23.2)
BAH	251.6	568.7	1169.3

Table 4 Comparison on profits of different out-sample periods

We can find that all of 3 sets of parameter can beat BAH strategy in different out-sample periods, with (14pt, 110d) performs the best. Then we want to know when is the enter point and exit point for the strategy of (14pt, 110d) to generate profits.

Parameter set	Purchase Day	Purchase Price	Sold Day	Sold Price	Profit
1. 2017-Apr-03 to 2018-Mar-29	2017-Apr-10	2352.9	2018-Jan-30	2854.2	501.3
2. 2016-Apr-06 to 2018-Mar-29	2016-Apr-06	2041.7	2018-Jan-30	2854.2	812.5
3. 2013-Jan-02 to 2018-Mar-29	2013-Jan-02	1441.1	2018-Jan-30	2854.2	1413.1

Table 5 Comparison on transaction date of different out-sample periods

Interestingly, the sell signal only presents once on 2018-Jan-30 no matter which set of parameters we choose. The different profits are solely due to various purchasing prices. It reinforces our

previous suggestion to separate buy threshold from sell threshold as the profits would be more sensitive to the purchasing point compared with the selling point.

(4) Other Considerations

In this part, we consider the factors we may face if deploying this strategy in the real-life trading.

First, the long-term holding cannot create sustain cash inflow. We are facing the liquidity risk if the investors withdraw the funding during crisis. It will create a huge loss if the sell signal is not triggered but we are forced to close our position at a very low selling price.

Second, the selection of parameters may involve subjective judgement. The profits are highly dependent on the chosen threshold and holding period. Even a slightly change in the threshold, a buy or sell signal will not be generated, thus the profit would be significantly different. So we need to be very careful when selecting the parameters.

Last but not least, we consider the transaction cost to buy and sell futures as well as roll over the future at the month end. However, it is considered to be immaterial as we trade very infrequently and the transaction cost for futures is minimal.

VI. Conclusion

As shown in our data and analysis, the strategy of buying SPX futures when VIX index goes above a threshold and closing the position when VIX fails to sustain above the threshold for a certain number of days can beat the buy-and-hold strategy for some combinations of the threshold and holding period parameters. When choosing the parameter, not only do we need to consider the performance in back-testing and out-sample period, we also need to take factors such as transaction cost and liquidity need into account, and it is possible to improve the strategy by adding another threshold for VIX index to determine the timing of closing the position.

VII. Reference

Tom McClellan. 2017. Correlation Between VIX and SP500. Available at: <http://stockcharts.com/articles/tac/2017/06/tom-mcclellan-correlation-between-vix--and-sp500.html>. [Accessed 6 May 2018].

Daniel Myers. 2018. Contrarian Investing. Available at: <https://www.investopedia.com/articles/financial-theory/08/contrarian-investing.asp>. [Accessed 6 May 2018].

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VIII. Appendix

```
setwd("~/C:\\temp")
d<-read.csv("vix_1.csv",header=TRUE)
plot(d$Vopen, main="VIX")
density(d$Vopen)
#bah as reference
bah=d$SPXopen[length(d$SPXopen)]-d$SPXopen[1]
bah

#buy SPX at open when VIX open is bigger than x
#this indicates buy when the market is volatile because
#investor doesn't like volatility and price might go down
#keep holding through the volatile day and sell when it is
#less volatile- this is indicated by y days that VIX
#hasn't been above x

#trading strategy backtester, return profit vector
sim<-function(threshold,hp) {
  signal<-(d$Vopen>threshold)
  fh=hp
  fc=0
  profit = vector('integer')
  tx = vector('integer')
  bp=0
  hold=FALSE
  for (i in 1:length(signal)) {
    profit[i]=0
    tx[i]=0
    #buy case if signal is true
    if(signal[i] && !hold) {
```

```

    hold=TRUE
    bp=d$SPXopen[i]
    tx[i]=-bp
    fc=0 #reset counter
  }
  if(signal[i] && hold) {
    #do nothing
    fc=0
  }
  if(!signal[i] && hold && fc < fh) {
    fc=fc+1
  }
  if(!signal[i] && hold && fc >= fh) {
    #sell as held too long over "quiet" time
    profit[i]=d$SPXopen[i]-bp
    tx[i]=d$SPXopen[i]
    hold=FALSE
    bp=0
    fc=0
  }
}
profit
}
#to find the i that is appropriate
result=vector();
for(i in 1:100) {
  result[i]=sum(sim(i,20))
}
result
#to find the j that is appropriate

```

```

result=vector();
maxresult=vector();
for(j in 1:30) {
  result=vector();
  for(i in 1:70) {
    result[i]=sum(sim(i,j*10))
  }
  maxresult[j]=max(result)
}
result
maxresult
bah
# Plot VIX and SPX for the whole period
plot(d$Vopen,type='l',col="red",ylab = 'index point',xlab=")
par(new=TRUE)
plot(d$SPXopen,axes=FALSE,type='l',col="green" ,ylab="",xlab=")
axis(side=4, at = pretty(range(d$SPXopen)))
mtext("z", side=4, line=3)
legend('bottomleft', legend=c("VIX", "SPX"), col=c("red", "green"),lty=1:2, cex=1.0)
# Trading performance in out-sample period (1 year)
test<-d[2330:2579,]
#bah as reference
baht<- test$SPXopen[length(test$SPXopen)]-test$SPXopen[1]
baht
# Plot VIX and SPX for the out-sample period in 2018
plot(test$Vopen,type='l',col="red" ,ylab = 'index point',xlab=")
par(new=TRUE)
plot(test$SPXopen,axes=FALSE,type='l',col="green" ,ylab="",xlab=")
axis(side=4, at = pretty(range(test$SPXopen)))
mtext("z", side=4, line=3)

```

```
legend('bottomleft', legend=c("VIX", "SPX"), col=c("red", "green"),lty=1:2, cex=1.0)
```

```
#Non-consecutive Counting Strategy Count of Holding Period
```

```
sim<-function(threshold,hp) {  
  signal<-(test$Vopen>threshold)  
  fh=hp  
  hd= vector('integer')  
  profit = vector('integer')  
  tx = vector('integer')  
  bp=0  
  fc=0  
  hold=FALSE  
  for (i in 1:length(signal)) {  
    profit[i]=0  
    tx[i]=0  
    hd[i]=0  
    #buy case if signal is true  
    if(signal[i] && !hold) {  
      hold=TRUE  
      bp=test$SPXopen[i]  
      tx[i]=-bp  
      fc=0 #reset counter  
      hd[i]= fc  
    }  
    if(signal[i] && hold) {  
      #do nothing  
      fc=0  
      hd[i]= hd[i]  
    }  
    if(!signal[i] && hold && fc < fh) {
```

```

    fc=fc+1
    hd[i]=fc
}
if(!signal[i] && hold && fc >= fh) {
    #sell as held too long over "quiet" time
    profit[i]=test$SPXopen[i]-bp
    tx[i]=test$SPXopen[i]
    hd[i]=fc
    hold=FALSE
    bp=0
    fc=0
}
}
hd
}
#(1)13,40
result=vector();
result=sim(13,40)
result
test[150,]
test[200,]
#(2)14,110
result=vector();
result=sim(14,110)
result
#(3)17,300
result=vector();
result=sim(17,300)
result

```